Anurag Upadhyaya

+91-9717515228 anuragu901@gmail.com LinkedIn Github

SUMMARY

Computer Science undergraduate with a focus on building scalable distributed systems and fault-tolerant backend architectures. Experienced in designing and implementing multi-tiered applications using C++, Node.js, and Next.js, with a proven ability to manage high-volume concurrent user sessions and optimize complex data query systems. Proven ability to solve complex, broadly defined problems by implementing data-driven optimizations that enhance system speed and scalability while maintaining a strict focus on operational excellence

EDUCATION

Noida Institute of Engineering and Technology **2022 – 2026**
B.Tech - Computer Science & Business System - ***CGPA - 8.17*** *Greater Noida, UP*

Kendriya Vidyalaya JNU Campus **2021**
Class 12th - ***Percentage - 88%*** *New Delhi, Delhi*

Kendriya Vidyalaya **2019**
10th - ***Percentage - 92.4%*** *New Delhi, Delhi*

PROJECTS

Next.js Full-Stack Auth App | Next.js 15, React, TypeScript, Express.js, MongoDB **August 2025**

- Orchestrated a resilient, stateless authentication protocol for 10,000+ concurrent sessions using JSON Web Tokens and HTTP-only cookies, mitigating common web vulnerabilities like XSS.
- Constructed a type-safe API contract with 5+ server-side route handlers in Next.js, leveraging TypeScript to eliminate a class of runtime errors and guarantee data consistency.
- Devised a unified gatekeeping middleware to programmatically shield over 4 protected application routes, centralizing authorization logic and enhancing system maintainability.
- Modeled an extensible user database schema with Mongoose, enforcing 8+ programmatic validation rules to fortify data integrity at the persistence layer and prevent malformed entries.
- Instituted a dual-layer validation strategy, leveraging real-time client-side checks and Mongoose schemas to enforce data integrity. This preemptively blocks over 95% of malformed submissions, providing instantaneous user feedback and reducing server-side error handling.

VidTube Backend system | Node.js, Express.js, MongoDB, Cloudinary, Multer **May 2025**

- Engineered a scalable backend API capable of handling **10,000+** concurrent users by implementing a secure JWT-based authentication system with refresh/access token rotation.
- Integrated a cloud-based media processing pipeline with Cloudinary and Multer, enabling efficient video and image uploads and reducing server load by **100%** for media-related tasks.
- Developed a robust subscription and social interaction model (likes, comments) designed to manage over **1 Million** relationships and interactions efficiently.
- Optimized database query performance by **40%** using advanced MongoDB aggregation pipelines for complex data retrieval, such as user watch history and channel analytics.

TECHNICAL SKILLS

Programming Languages: C, C++, Python

Backend & Systems: Node.js, Express.js, RESTful APIs, Distributed Systems, Scalable Architecture, JWT Authentication/Authorization.

Databases & Storage: MongoDB (Advanced Aggregation Pipelines), SQL, Mongoose ODM, Cloudinary, Multer.

Computer Fundamentals: Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, Database Management Systems

Tools & DevOps: Git, GitHub, Agile Project Management, Postman.

Soft Skills: Team Work, Adaptability, Leadership, Communication.

CODING PROFILE

- Solved **300+** Data Structures & Algorithms based questions on the mentioned coding platforms
- **LeetCode**
- **GeeksForGeeks**
- **CodeChef**